

# Progressive Addition Lenses: History, Design, Wearer Satisfaction and Trends

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**Abstract:** Progressive addition lenses are currently the state of the art in multifocal correction for presbyopia. This paper discusses the trade-offs inherent in progressive lens designs; adaptation to and the measurement of progressive lenses; as well as current and future trends in the industry.

## Introduction

Presbyopia literally means 'Old eyes' and reflects a clinically significant loss of amplitude of accommodation, or ability to focus on objects at near distances. Everyone suffers a steady loss of accommodative capacity – from 16 diopters (D) at 6 years of age to about 6 D at 40 years and effectively none by 60 years. The loss of accommodation is the result of aging of the lens, tendons and elastic fibers in the eye [1] and is still poorly understood. Most near tasks require 2 to 4 D of accommodation (25 to 50 cm viewing distance), so soon after 40 years of age, accommodative capacity begins to fall short of task demands.

To be able to wear spectacles without the need to take them on and off, presbyopes need at least two different corrections, one for distance viewing and one for near viewing. A major advance in spectacle lenses came in the 19<sup>th</sup> century with the invention of the bifocal lens to aid presbyopic wearers. This allowed two different prescriptions to be contained in the same frame. The invention of the bifocal is usually attributed to Benjamin Franklin, but some bifocals may have existed previously [2]. Franklin at least managed to popularize their use. He described the reasons for and the making of his bifocals in a letter to an optician friend, George Whately, in 1784 [3]. Franklin appears to have simply taken a distance lens and a near lens and cut them both in half, putting the distance correction in the top half of a circular frame and the near correction in the bottom half. In this manner, Franklin's original bifocals were similar to today's "executive" bifocals, with a line running all the way across the lens.

Since the invention of lined bifocals, the method of manufacture, size and location of the segments, and number of segments have been modified to create today's designs. One of the most distracting problems for wearers of bifocals is the visible line dividing the near and distance sections.

Designed to overcome the problems of bifocals, the current state-of-the-art in multifocal lenses is the progressive addition lens (PAL). PALs have no visible lines, as there are no discontinuities in power on the surface of the lens. Instead, the power varies continuously through the intermediate or progressive zone between a stable distance zone and a near zone. This is achieved by continuously decreasing the radius of curvature on the front surface.

Patent activity in progressive lenses began as early as 1907 [4], but the first commercial PAL was the original Varilux lens launched in Europe in 1959. The first PAL launched in the US was the Omnifocal lens released in 1961. It is fair to say that the widespread up-take of PALs did not occur until after the original Varilux patent expired and a variety of PAL designs was introduced. Today, one in four lenses sold in the US are PALs [5].

PALs have several advantages over other forms of multifocals – no lines to interfere with cosmetics, no sudden power changes which cause image shifts, and corrected vision at all distances. These benefits come at the cost of increased peripheral astigmatism and distortions. It is impossible to create a progressive surface without optical errors in the periphery of the lens [6]. The typical layout of a PAL is demonstrated in Figure 1.

Figure 1 shows the levels of surface astigmatism in a PAL. Each change in shading is equivalent to 0.5D of astigmatism with the darkest area indicating between 1.5 and 2D.

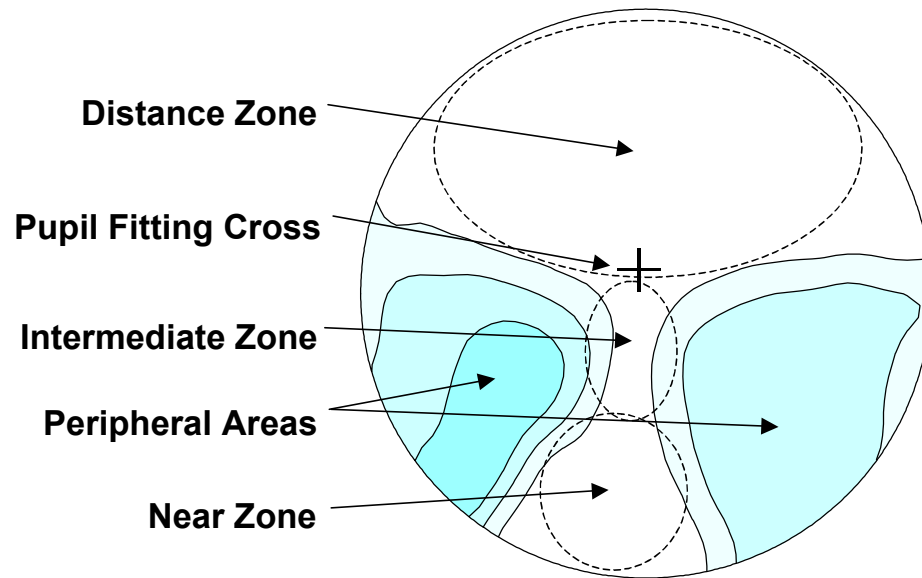


Fig. 1. Schematic plot of a progressive lens with zones labels.

There have been several prior reviews of progressive lenses [4, 6-10]. The purpose of the present article is to update these with current knowledge and to offer a different perspective.

### Trade-offs

Given that it is impossible to design a progressive surface without peripheral astigmatism or distortion, and that PALs have been on the market for more than 40 years, one might wonder why new PAL designs are introduced. Advances in the mathematics of progressive surfaces, design software, and milling technology have enabled PAL companies to reduce the peripheral distortions closer to the theoretical limits. Perhaps more importantly, designing a PAL is a matter of balancing a large number of mutually exclusive design features – in short, making trade-offs. Given the almost infinite combinations and permutations of designs, the variety of tasks for which the lenses will be used and advances in our understanding of the human visual system, there simply is no single right answer for all wearers and all situations. Changing work and leisure environments may also render previously acceptable designs inappropriate. Advances in understanding in any one of the above areas may inspire a new series of PAL designs.

Lens designers are able to alter zone sizes and move the locations of the distortions around the lens. However, these are not independent variables. Changing the size of one zone affects the size of the others and alters the levels of the distortions. For example, some wearers complain that they have to tilt their head back too far in order to be able to read text at close range. One solution is to shorten the length

of the progression zone, commonly called the corridor length. This might enable the wearer to read without tilting her head back as far, but it raises the levels of the peripheral distortion and blur which might cause more inconvenience than having to tilt one's head back to read. Alternatively, fitting the design higher may result in addition power or distortions impinging on distance vision

Other patients object to having to move their head from side to side to read because the size of the near zone is not wide enough. The lens designer can widen the near zone but, as in shortening the corridor, this causes other problems. As I will discuss later, wider is not always better in near zone width since it raises the levels of blur and distortion elsewhere in the lens. The task of finding the right balance is a difficult one and will always be open to investigation.

These preceding two examples are demonstrated in Figure 2. Surface astigmatism and surface mean power are shown for three different PAL designs: a "standard" lens, a short corridor lens and a wide near zone lens. All three surfaces were generated for the purposes of this paper. I attempted to only change one variable from the normal lens. These lenses are not products, have not been optimized, and are for illustrative purposes only.

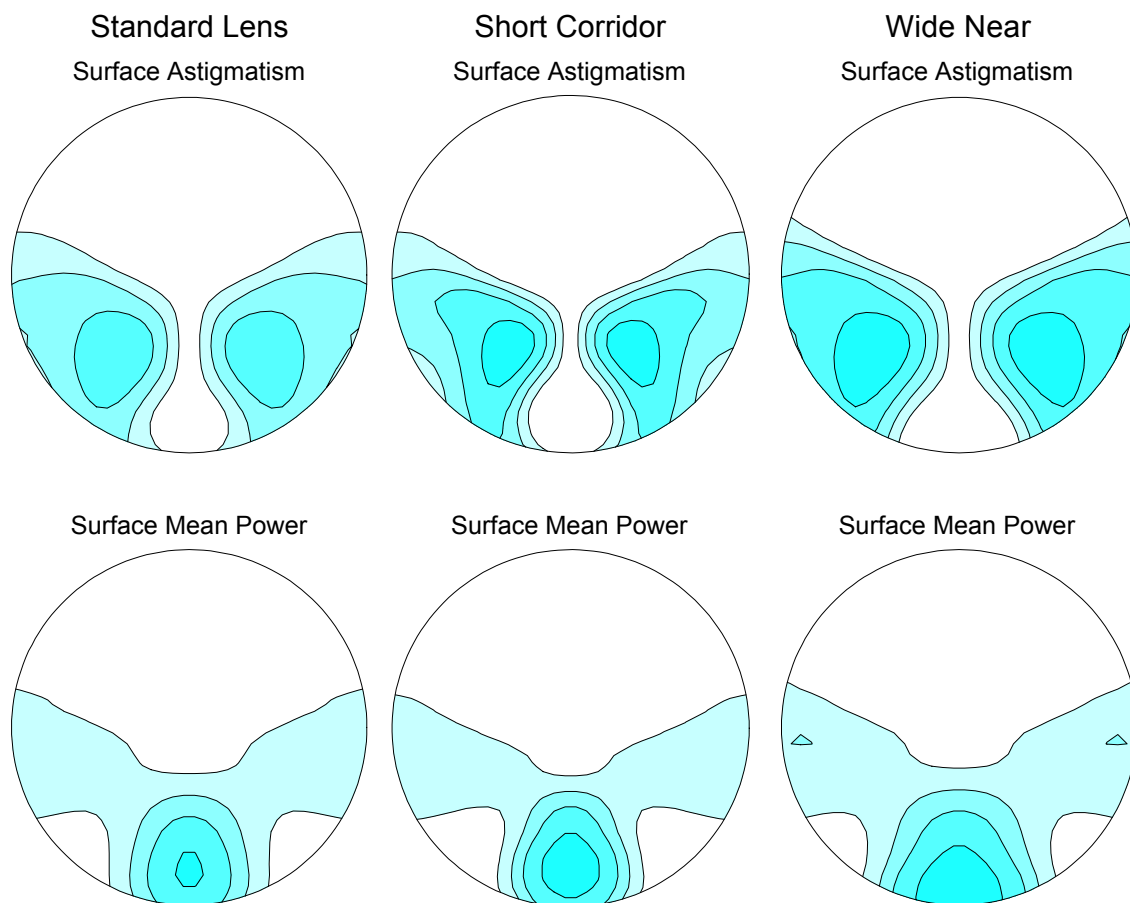


Fig. 2. Demonstration of the effects of shortening the corridor or enlarging the near zone of PALs

A general rule is that to minimize distortion you must distribute the changes in curvature of the lens over a larger area. This results in smaller zones of clear vision. Once zone sizes have been set, the lens designer still controls how blur and distortions are distributed in the peripheral zones. The designer

can try to minimize the peak of astigmatism, the gradient of astigmatism, the acceleration of astigmatism, or simply move the peak around.

One other point to keep in mind is that the patient's frame choice affects the blur and distortion they will see through the lens. The progressive lens plots shown have a diameter of 65 mm. This is much larger than most patients' frames. The smaller the frame, the more of the peripheral blur and distortion that will be edged off the lens. The problem with smaller frames is that the zones of clear vision are also cut down. If the frame gets too small, the full addition power of the lens will not be achieved, with the entire near zone edged off. To minimize distortion, patients should select the smallest frames that give them adequate zones of clear vision.

## **Wearer Trials**

One way that PAL manufacturers investigate how wearers like the trade-offs that they have made and get an idea of potential success in the marketplace, is through wearer trials. In wearer trials, subjects wear the lenses being studied and rate their performance. Wearer trials are necessary because no single surface or optical measurement of a lens has been found to conclusively predict how wearers will respond to PALs.

There are two main types of PAL wearer trials: acceptance and preference. In acceptance trials, subjects are given a PAL to wear and are later asked if they would like to keep the lens or return to what they were previously wearing. In preference trials, subjects are given two different new lenses and are asked which one they prefer. In general, preference trials give much less impressive numbers than acceptance trials. I will further break up the preference trials into two groups: PAL vs. Bifocal and PAL vs. PAL.

### ***Acceptance Trials***

Acceptance trials generally have two different measures of success. One is whether the subject is still wearing the lens being studied at the end of the trial (acceptance), and the other is satisfaction. The measure used to gauge satisfaction varies from trial to trial and care must be taken when interpreting these results. Two examples of satisfaction would be if the subject states they would buy the trial lenses or a satisfaction rating of 3 or better on a 5 point scale (1 being the worst score, 5 the best). Some examples of PAL acceptance trials are Schwartz and Schwartz (1967), Wittenberg (1978), Chapman (1978), Augsburger et al. (1984), Sullivan and Fowler (1989), Cho et al. (1991), Gresset (1991), Boroyan et al. (1995), Young and Borish (1994) and Kris (1999) [11-20].

The results of these acceptance trials are reasonably similar. This makes sense since most of the trials are funded by the PAL manufacturer and the studies are intended to convince practitioners to prescribe the lens. Several different lens designs were studied, but by far the most common was the Varilux 2. The average level of acceptance in these trials is 93% and varies between 86 and 99%. The satisfaction numbers tended to be lower, averaging 82% and ranging between 72 and 92%. This suggests that a significant number of patients will continue to wear a lens that they are unhappy wearing. Since these subjects only had their prior lenses to compare to, their happiness and acceptance of the PALs might have been influenced by how out of date their old prescription was. Had they all been comparing to up to date prescriptions satisfaction rates may have gone down.

The results of several trials suggest that some subjects have a higher probability of success in a new PAL. Previous PAL wearers [17, 19], those with a high distance prescription [15, 16] or lower add power [15], and those for whom the trial lens was their first multifocal prescription [14, 17] appeared more likely to succeed with a new PAL.

One interesting conclusion of the Young and Borish [20] study is that “Success is apparently influenced by fitting skills” of the practitioners. They base this on the finding that isolated practices had acceptance rates much lower than most of the others. A similar point was made in a very recent article in a trade magazine [5] in which the author concludes that proper fitting and patient education are even more important than the design.

### ***Preference vs. Bifocals***

A number of wearer trials [11, 21-24] pitted PALs against bifocal lenses in an effort to get practitioners to prescribe PALs as their multifocal of choice. Many practitioners had good success rates with bifocals and were reluctant to risk patient unhappiness by prescribing a new lens type unless it was asked for.

In all of the wearer trials of this type, PALs are favored over bifocal corrections. An interesting trend is that the preference numbers in favor of PALs increase in more recent studies. In the earliest study, preference was only 52% for PALs, while the most recent studies find preferences over 90%. This suggests that PAL designs are improving, as it is unlikely that bifocals have gotten worse. It is also possible the increasing popularity and prevalence of PALs and increased practitioner enthusiasm has meant that bifocal wearers are more motivated to wear PALs. Common reasons cited for the preference for PALs are better cosmetics (due to the absence of a line), better intermediate vision, no image jump and more comfortable vision. The main reason cited by those who selected bifocals was a wider near field of view.

### ***Preference vs. Other PALs***

Not surprisingly, in all the published wearer trial studies that I have seen comparing one PAL to other PAL designs [25-30] the design of the company sponsoring the research showed the highest preference ratings. From this one can conclude that either new PAL designs are always better than all old designs, or more likely, unfavorable studies are not published. The one study [30] that did not name the lenses used but described their qualities had the closest preference scores, 52 to 48%. This study seems to indicate that different design philosophies can be equally preferred.

One confounding variable in all preference studies of this type is the prior lens worn by the subjects. Since subjects have presumably fully adapted to the particular blur and distortion pattern of their old lenses, which are often PALs, any lens that is similar to their old lens is likely to have an advantage over all other PALs. Selectively recruiting subjects with prior experience wearing lenses similar to the manufacturer’s lens is one way that studies can be biased toward favorable results.

One interesting finding in the Wittenberg et al. (1989) [25] study is that males seemed to prefer the designs with lower gradients of astigmatism compared to females. The gradient of astigmatism refers to the rate at which the astigmatism increases at the distance and near boundaries. Lenses with low gradients are often called “soft” designs, while lenses with high gradients are called “hard” designs.

Another study [31] worth mentioning here combined aspects of all three types of wearer trial discussed. Acceptance data, preference data against bifocals and preference data between two PALs was collected. Most of the results closely matched those presented earlier, but some interesting conclusions were reached. Although there was a clear preference for one of the PAL designs over the other in the forced choice, when asked about the strength of the preference, subjects did not usually state a strong preference. The authors conclude, “the impression was conveyed that had only one type been offered, the total acceptance would still have been comparable.” They also note that the “most satisfactory response was associated with those practitioners whose own enthusiasm for variable focus lenses was greatest.” For acceptance of progressive lenses, one should not underestimate the influence of the dispenser.

### ***Preston Study***

A recent Ph.D. thesis written by Julie Preston [10] is especially interesting because it contained two wearer trials in which lenses with carefully controlled differences were compared. This differs from most previous comparison studies in which released products were compared which varied in an uncontrolled manner. None of the designs were products and except for the variable being studied the lenses were kept as close as possible. One trial compared lenses with short, medium and long corridors and the second compared lenses with narrow, medium and wide near zones. The medium lenses in both trials were the same. See Fig. 2 for an idea of the differences between the lenses. Each lens was evaluated immediately and again after one week of wear. Lens order was randomized. All subjects were prior progressive lens wearers.

When choosing between lenses of different corridor lengths, the results were not very clear-cut. The winning design depended on the measure used. The short corridor had the highest average satisfaction ratings, the medium corridor lens was chosen most often to keep and the long corridor lens was the easiest to adapt to. One interesting trend was that the score for the long corridor lens fell during the one week of wear. The initial impression of the lens was more favorable than after a week of wear. As time went on subjects seemed to appreciate the improved ergonomics of the shorter corridor design. They were willing to give up the “softer” design with the lower levels of peripheral astigmatism and distortion.

In the comparison between lenses of different near zone sizes the results were much clearer. The medium width design was the favored design, while the wide near zone design was the least favorite by far. The gap between the wide design and the other lenses was even greater after one week of wear compared to the initial impressions. In this case, it was the “harder” lens that suffered with time. Reading ergonomics may have been better with the wider near zone, but subjects seemed unwilling to give up the advantages of the narrower designs. It seems that once a near zone is wide enough, making it any wider hurts the acceptance of the lens. One possible problem with this finding has to do with the actual wide lens design used. It did not widen until fairly far down the lens, and may only have widened in a part of the lens wearers do not like to use.

These wearer trials also pointed to some interesting trends. It appears that newer progressive lens wearers (<4 years of wear) preferred the longer corridor length and the narrower near zone width. More experienced wearers, however, preferred the medium near zone width and either the short or medium corridor length. The more experienced wearers seem better able to tolerate peripheral blur and distortion in order to get better ergonomics.

The studies also pointed to possible gender differences, similar to those suggested by the Wittenberg (1989) study [25]. Men preferred the narrow design while women preferred the medium design. Women preferred the short corridor (but found all acceptable) while men preferred the medium length and rejected the short corridor. One confound in the study is that the men in the study tended to have less experience than the women so the findings about experience and gender are not independent.

### ***Objective Measures Wearer Trials***

Another type of wearer trial that is just starting to appear is one in which objective measures are used to compare the performance of different lens designs. These studies are capable of demonstrating advantages of one lens design over another, but the conclusions might only apply to the specific set of circumstances tested. One such study found that aviators wearing a PAL correction were significantly faster locating static targets on the cockpit instrument panel compared to the same aviators wearing a bifocal correction [32]. Other studies have found differences in head and eye movements and time to perform a reading task when PALs are compared to single vision lenses or other PALs [33, 34].

## Adaptation

One consistently recurring issue for PALs and for multifocals in general is adaptation. A portion of presbyopes, albeit small, fail to adapt to their lenses. PAL manufacturers would like to minimize non-adapts, and much of the current marketing literature promises “easy adaptation.” In fact, minimizing non-adapts may be more important to manufacturers than maximizing the happiness of the successful wearers. Given two PAL designs, one may be released because it promises fewer returns instead of getting the highest satisfaction scores. The two variables do not necessarily correlate exactly when comparing “good” designs. Non-adapts cost patients, practitioners and manufacturers money and can damage the reputation of the lens, while customers who are not as happy as they could be are still satisfied enough to keep the lenses.

Adaptation is a complex area that has many aspects. The adaptation phenomena that most think of immediately is “swim”, an image distortion in the periphery of a PAL that is apparent in dynamic situations. Swim can be visually disturbing and can cause physical distress. Patients often report dizziness, headaches and nausea if swim is perceived.

There are other components to adaptation as well. The wearer must adapt to the ergonomics of the lens. Every lens has different zone sizes and locations, and the wearer must learn how to position their head and eyes to see objects at all distances clearly. Ergonomic problems can lead to frustration with the quality of vision and muscle aches as neck, shoulder, back and eye muscles work to adopt a posture that matches the lens zone with the task position. Another component is static distortion, where straight lines appear to be bent, which some wearers find quite disturbing. There are also vague “comfort” issues, which may result from unyoked prism differences between the left and right lenses causing ocular motor stress.

Subjects appear to be able to adapt to complex prism fields [35-38] and it even appears that patients can learn multiple sets of lens adaptations [38], but not all subjects respond the same way. One patient may be able to switch between several different lenses at will, while another may have great difficulty any time they change anything about their lenses.

The wearer trials in the Preston thesis [10] indicated that preferences for zone sizes and locations seem to change with time. This implies that for optimal happiness with a lens, it might be worth a painful adaptation process in order to attain better lens ergonomics. The advice commonly given out by dispensers with new lenses to “give the lens two weeks before giving up” might be sound advice.

The advice is also shown to be sound in a recent unpublished wearer trial conducted by SOLA Optical. Each wearer participated in two separate trials in which they compared two new PAL designs with their own normal PAL. Each of the new lenses was worn for one week. One of the new lenses in each of the two trials was the same, although the subjects did not know this. At the end of each trial, subjects were asked to rate their happiness with each of the lenses, including their own. Figure 3 shows how the number of subjects unhappy with the new lens and their old lens changes after two weeks. After one week of wear about 20% of wearers were dissatisfied with the new lens while after the second week of wear that number dropped to about 7%, which was almost exactly the same as the number unhappy with their own lenses. This apparent improvement is especially impressive given that there was up to a one-month period between exposures to the new PAL. This lends even more support to the theory that wearers can store adaptations to several lenses at once for reasonable periods.

Another thing to remember about adaptation is that some wearers will always fail to adapt. However, if a practice’s failure rate is higher than an acceptable level over a period of about 100 wearers the practice’s fitting, especially PD measurements and fitting height, should be checked, patient education should be looked at, and finally the choice of lens design should be rethought. It is, however,

difficult to determine if the non-adapt rate is too high after only a couple of wearers. Barring a major problem along the lines of those mentioned above, non-adapts should be random. If there is a 4% non-adapt rate, it does not mean that one in every twenty-five patients will fail to adapt. Probabilities work out in the end but in the short term large anomalies are common. It is quite possible for three of the first ten to fail. This is demonstrated in Table 1.

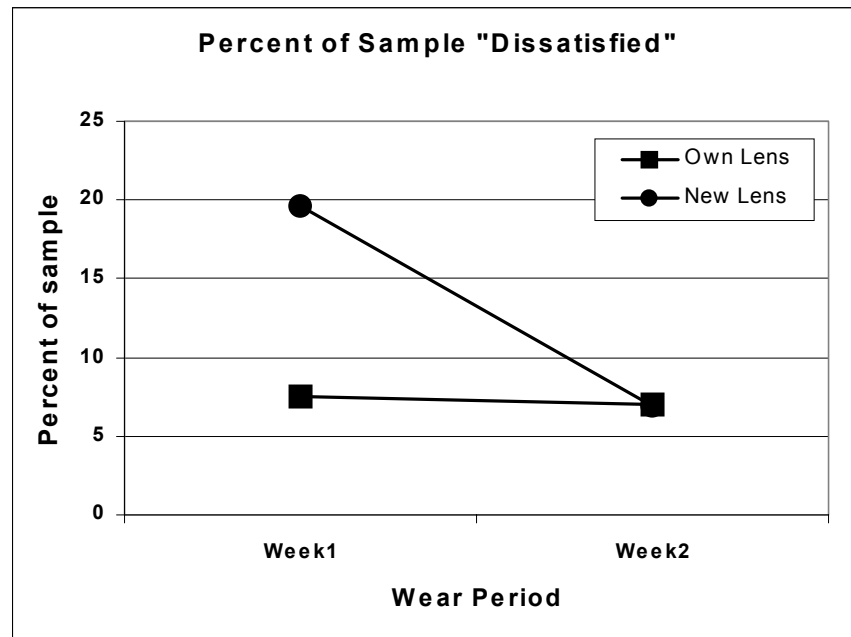


Fig. 3. Shows the percentage of subjects unhappy with their own lens and a new lens after one week and two weeks of wear. Subjects were not told that the new lens was the same in each period.

Table 1 models twenty-five practices all of which have a 4% failure rate with PALs and shows non-adapts (a 1 in the box) in the first ten wearers as well as the total in the first one hundred (top row.) The data in the table were randomly generated in Microsoft Excel, and the author can be contacted for a copy of the sheet. There are several interesting things to notice about the table. Although most of the practices had no failures in the first ten, one had three and two had one. It is easy to imagine these practices assuming that their new progressive lens was no good, quitting their use of the new PAL and telling their colleagues that the lens is no good. The practice with 3 failures in the first 10 then went on to have only two more in the next ninety. It is dangerous to make decisions based on small numbers of wearers. Also, note that practice 2, which had no non-adapts in the first ten, had the most in the first hundred. After being initially satisfied, they may conclude that something about the lens must have changed. In general, it is common for practitioners to dismiss failures with favorite designs as patient related, forgetting the incident quickly, but blame the lens design immediately if they have trouble with a new one.



Table 1. Randomly generated model of non-adapts in the first 10 and 100 patients in 25 practices with a 4% intrinsic failure rate. A 1 in a box indicates a non-adapt.

| Practice No.                | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |
|-----------------------------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| Total Failures in first 100 | 5 | 7 | 1 | 5 | 5 | 5 | 6 | 6 | 5 | 2  | 4  | 2  | 4  | 2  | 3  | 7  | 3  | 2  | 3  | 2  | 1  | 5  | 6  | 2  | 2  |
| Patient #                   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 1                           |   |   |   |   |   |   |   |   | 1 |    |    |    |    |    |    |    |    |    |    |    |    |    | 1  |    |    |
| 2                           |   |   |   | 1 |   |   |   | 1 |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    | 1  |    |
| 3                           |   |   |   |   | 1 |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 4                           |   |   |   |   |   |   |   |   | 1 |    |    |    |    |    |    | 1  |    |    |    |    |    |    |    |    |    |
| 5                           |   |   | 1 |   |   |   |   | 1 |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 6                           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    | 1  |    |    |    |    |    |    |
| 7                           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 8                           |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
| 9                           |   |   |   |   | 1 |   |   |   | 1 |    |    |    |    |    |    |    |    |    |    |    |    | 1  |    |    |    |
| 10                          |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |

## Analysis Issues

One area that is not very consistent within the PAL industry is how to best analyze lenses. There are many issues here, from whether the basic analysis should be surface quantities or optical quantities to what is the best method to measure corridor length and zone sizes. Manufacturers like to be able to say that their lens is the best, so the method that they use to “objectively” compare PALs is likely to be the one that makes their lens look the best. There is little incentive for manufacturers to settle on one standard method to analyze a lens.

Contour plots of various lens quantities, usually astigmatism and mean power, are often included in product marketing literature. Traditionally these have been plots of front surface characteristics or of lenses mapped with a focimeter. Neither of these techniques gives an accurate picture of how the lens will perform when worn [39]. A recent trend is to use plots based on a ray trace of a complete lens in front of a model eye or where the optical mapping is based on eye point performance [40]. Accurate ray tracing has only become a possibility with fast modern computers and allows a lens designer to determine the as-worn performance of a potential lens that only exists as a design file on a computer. Once a surface is stored in a manner that the ray-tracing program can use, an almost unlimited number of analyses can be performed. Ray traced analysis or eye point mapping offer a marked improvement over older methods of analysis.

Figure 4 demonstrates some of the capabilities of modern ray tracing software. All of the plots are based on the same PAL surface, the standard surface from Figure 2. The top three contour plots display the surface attributes of astigmatism, mean power and root mean squared (RMS) power. The RMS power here is an average dioptric focal error, is numerically similar to Sloane Blur and is a good method to characterize expected levels of perceived blur. It is also equivalent to the square root of the sum of the squares of the three Zernike fitting coefficients for the wavefront error associated with defocus and astigmatism. If the principal powers of the total power error are E1 and E2 then:

$$RMSPowerError = \sqrt{\frac{E1^2 + E2^2}{2}} \equiv \sqrt{\bar{P}^2 + (cyl/2)^2} \quad (1)$$

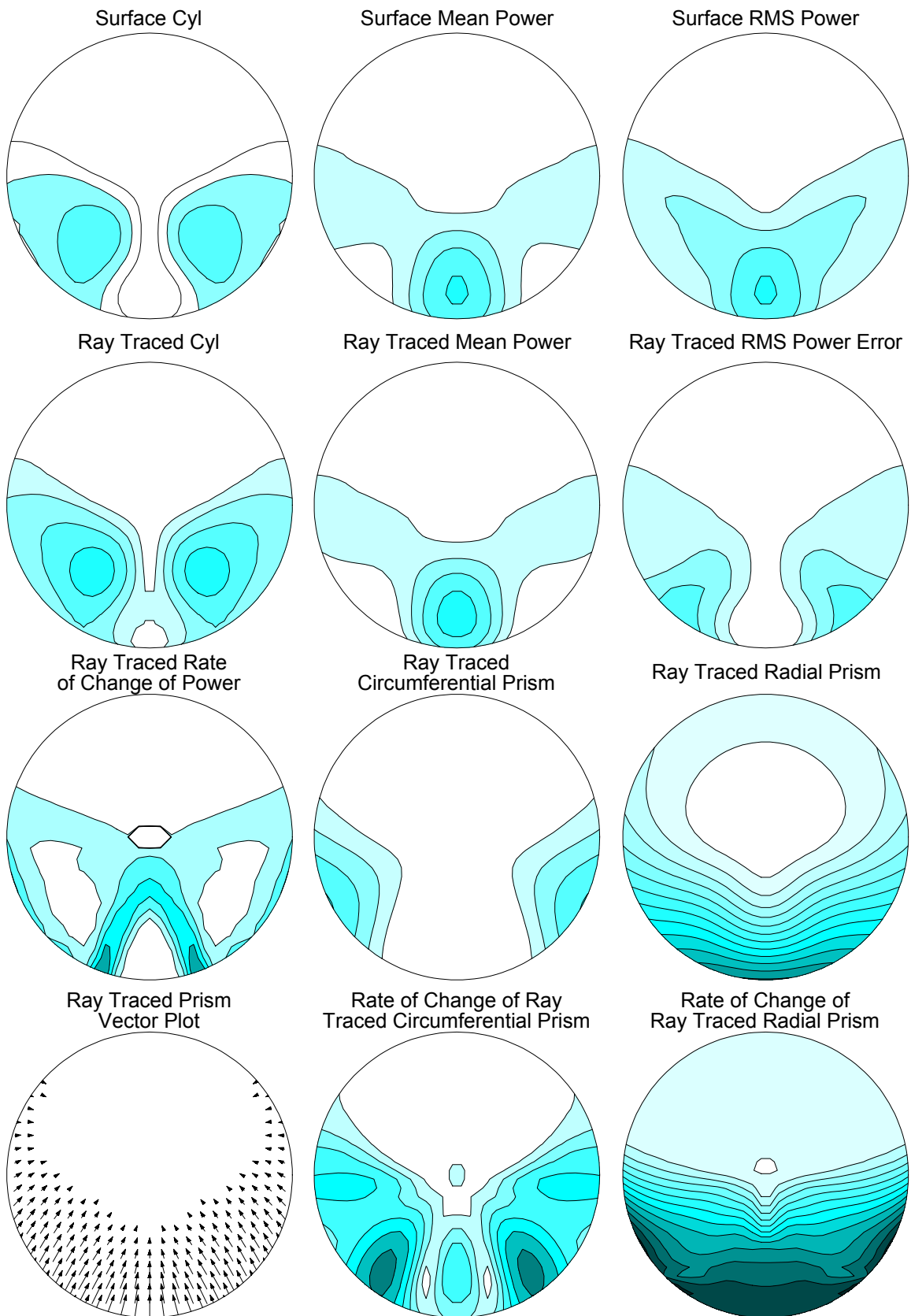


Fig. 4. Surface and ray traced plots demonstrating different characteristics of the same lens.

where  $P$  is the mean power error and  $cyl$  is the magnitude of the astigmatism. The power of the base curve of the surface is subtracted off all power values. For more detail about this quantity, see Raasch (1995) [41]. What is in this paper called RMS Power Error Raasch refers to as the Vector Length.

The second row of three plots displays the ray traced equivalents of the top three surface plots. Note the higher levels of astigmatism in the corridor and the higher peak astigmatism of the ray traced plot. The reason that the RMS Power plots look so different is that in the ray traced version an object field that varies from infinity at the top to 40cm in the near portion of the lens was used and the eye was given 0.5D of accommodation implying the ability to negate minus power errors up to 0.5D. The next six plots demonstrate measures that might be useful in determining the distortions seen by a wearer. They are also included to demonstrate the power of modern ray tracing software. There is, at present, no lens analysis that reliably predicts the distortions seen in a PAL by wearers.

Once accurate representations of a lens are obtained, the question then becomes which lens is better, which has the widest near zone, which has the shortest corridor, etc. The first question is still impossible to answer, as it is quite subjective. A designer, however, might be able to determine which lens promises the best vision for a well-defined subset of the population, and this might be determined by the answers to the later questions.

A question such as “How wide is the near zone” sounds quite simple but actually involves some complicated issues. One might wonder what determines the limits of clear vision for the near zone. Should it be a certain level of astigmatism, a specific RMS power error, or maybe the level of one of these quantities multiplied by the rate of change? One suggestion in the literature, based on eye-tracking studies, has been to use 1D of astigmatism as the limit of clear vision for the near zone [42]. Once a level is determined, the vertical position on the lens at which the width is measured becomes important. A lens that is very wide down low might actually be quite narrow where people are likely to be looking when reading; a short corridor lens should be measured higher than a long corridor lens. Possible rules for determining the height at which width measurements are taken include: a fixed distance below the fitting cross, the point at which 95% of the add is attained, and 3mm below the point at which 90% of the add is attained. A rule similar to the last one is probably a good candidate. Another possibility for determining near zone size is to use an area rather than a width. It is possible using ray tracing software to determine the percentage of an open magazine, or any other page size, at 40cm that can be seen with less than 0.5D of blur. This could ultimately give the best indication of the utility of a particular lens’ near zone. The same difficulties apply to the intermediate zone and distance zone as well.

The determination of the corridor length is also fraught with problems. Different manufacturers use different definitions. Possibilities include the distance from 5% to 95% of the nominal add, from the fitting cross to 90% of the nominal add, and the distance from the fitting cross to the nominal add minus 0.25D. There is little reason to choose one of these methods over another.

A recent series of patent applications from Essilor indicate some ways of thinking about these problems [43-45]. All three define the near zone width as the horizontal distance between the surface astigmatism contours equal to  $\frac{1}{2}$  the add power (eg. 1.25D of surface astigmatism for a 2.50D add design). This sets a tougher standard for lower add designs and will keep near zone widths more consistent between lenses of different adds.

One of the patents [44] introduces a dimensionless measure for the balance between the maximum slope of the power profile along the eye path and the maximum value of peripheral astigmatism within the foveal region of the lens, which is defined as a disk of 20mm radius centered on the geometric center of the lens. The measure is:

$$M = \frac{C_{\max}}{dP_{\text{mer}}} \quad (2)$$

where  $C_{\max}$  is the maximum value of the astigmatism,  $P_{\text{mer}}$  is the maximum value of the slope of the mean power along the principle meridian, and  $d$  is the radius at which this max slope is reached. This measure attempts to score lenses for their astigmatism removing the effects of the maximum slope of the power profile, which usually varies inversely with corridor length.

Another measure of near zone width is also introduced [44, 45]. It is an angular measure formed by radial lines originating at the geometric center of the lens and intersecting the 20mm radius circle at the point where the  $\frac{1}{2}$  add surface astigmatism contours intersect this circle. A possible improvement would be to measure the maximum angle that can be formed by radial lines starting from the fitting cross and touching the 0.5D ray traced RMS error contours for a 40cm object distance. For adds less than 2.50D, the eye would be given enough accommodation so that the accommodation and the add of the lens total 2.50D. This improvement would reward lenses with short corridors and those whose astigmatism contours open up high on the lens. There are no “right” answers here. Different criteria will reward different design types. The important thing is to try to understand the relationship between these measures and the wearer’s perception of the design’s performance.

## Recent and Future Trends

Manufacturers are constantly working to improve the design of PALs and new lenses are regularly released. Several trends can be discerned from recent PAL activity.

One trend is towards niche lenses, either task specific or frame specific. Task specific PALs such as SOLA Access, Hoya Desk, AO Technica and Varilux Readables, which are optimized for desk and computer situations, have been around for several years. These lenses, as typically prescribed, emphasize intermediate and near vision at the expense of distant vision. Several published studies [46-48], as well as unpublished studies done by SOLA, demonstrate that patients prefer this type of correction by large margins for near and intermediate work, yet they have never been large sellers. One reason is that they need to be purchased in addition to a standard PAL by all but emmetropic wearers who only wear lenses part time. This can add considerable expense. As more and more studies demonstrate the efficiency and wearer happiness benefits of these lenses, their use should increase.

A niche lens that has been very successful is AO Compact. This is a PAL designed to fit in small modern frames. It has a very short corridor, which increases the peripheral astigmatism, but since it is designed to fit in small frames much of the area of increased peripheral astigmatism is edged off the lens during fitting. Prior to the release of Compact many PAL purchasers had to choose larger, less fashionable frames in order for their lenses to achieve their full add. Compact managed to please a large number of PAL wearers who were unhappy with the choices previously offered. One advantage Compact had was that no second pair of spectacles was needed.

There is the potential for many more task/activity specific niche lenses. Some possibilities are – a driving PAL, an outdoor PAL, a golfing PAL, etc. Since progressive lenses all make trade-offs, no one lens is ideal for all situations. A PAL designed for specific tasks or activities has the potential to function much better in those situations. One very beneficial element of these for the PAL industry is that they have the potential to increase the PAL market size without increasing the number of PAL wearers. If every PAL wearer were to buy one general-purpose lens and one niche lens the PAL market would double.

A second recent trend is towards using customized back surfaces on PALs to improve the optics. This has only recently become viable in with the development of sophisticated computer controlled mills

and lathes. Both Rodenstock and Zeiss currently sell lenses on which they generate an atoric back surface designed to minimize the peripheral astigmatism. The problems with these lenses are that they take a long time to get to the consumer from the time of ordering, they are quite expensive, and they only make a real improvement in the optics for patients with high cylindrical components to their prescription. Even then, the optics are improved the most in a part of the lens that subjects seldom look through for critical vision. Rodenstock is looking to take this one step further by generating an atoric back surface on the lens that takes Listing's Law, which predicts the torsional position of the eye for every gaze position, into account by changing the axis of the cylinder at every position on the lens [49]. This lens would have the same problems as the current lenses and might cause additional swim problems due to the changing cylinder axis. As lens generating technology improves, providing quick and inexpensive custom back surfaces, this type of back surface might become much more common.

The trend towards customization has also taken place on the front surface of the lens in the PAL design. This customization is not for the individual wearer, but rather for wearers of the same add or similar prescription. It used to be that a 2.50D add PAL looked very similar to a scaled up version of the 1.00D add PAL in the same line of PALs, and a myopic base curve had the same design as a hyperopic base curve. Now many manufacturers including SOLA, Essilor and Rodenstock have determined that wearers with different levels of reserve accommodation and ametropias may behave quite differently or have different visual needs. The corridor length, zone size, corridor inset profile, power profile and pattern of peripheral astigmatism might all vary with the add power and the base curve of the lens. To get all of this right takes a very thorough understanding of wearer behavior and the optics of the situation. The increasing use of vision scientists by PAL companies, as well as modern optics software, make this customization possible and effective.

Another trend is the movement towards casting or molding PALs to Rx, or generating and polishing both the front and the back surface of the progressive lens. Casting or molding to Rx involves making a finished lens, with the cylinder axis correct, based on a patient's prescription. Most manufacturers sell PALs as semi-finished lenses where a separate lab generates a back surface based on the patient's prescription. If a manufacturer were able to accurately, efficiently and quickly make finished lenses to a patient's prescription they could sell finished PALs directly to the dispenser. This would cut out one potential source of error in filling prescriptions – the generation of the back surface. This "just in time" manufacturing would also reduce the inventory that a manufacturer had to hold. Paired with an express delivery company, the lenses could be delivered in one or two days. This type of manufacturing would not be easy, might have very high start-up costs, and would have to solve problems with edge and center thicknesses especially for plus lenses.

A process that generates both the front and back surfaces of a PAL from a cylinder of plastic might also have advantages. Inventory would be limited to stocking uncut blanks that could be cut to any design and prescription. Given further advances in PAL design software, the PAL design generated could be customized to the actual wearer based on questionnaire responses or some other measure. This method of delivery does currently have problems. The equipment needed to generate the surfaces is very expensive, the process is slow, there is currently no fast and reliable way to polish plastic progressive surfaces, and the lens design software is not far enough along. At least in the short-term, lenses that come from a system like this are likely to be very expensive items used for prototyping a design that will later be cast or molded.

## Summary

Progressive addition lenses represent the current state of the art in multifocal corrections for presbyopia. They currently account for about one out of four lenses purchased, and this is likely to rise as they become more accepted and the presbyopic population continues to grow [50]. Despite being on the market for more than forty years, new and improved PALs are consistently released.

The improved PAL designs come about due to greater understanding of progressive surfaces, refined manufacturing techniques, better design software, studying wearer ergonomics and determining what measurable quantities about PALs lead to wearer acceptance and happiness. These improvements can be used to ameliorate the sometimes antagonistic qualities of wearer adaptation and happiness. Much of the necessary information about wearers can be learned from carefully controlled wearer trials. However, most of the published wearer trials fail to tell us much about what makes a good PAL. They tend to compare designs of two or more competitors that vary in too many ways to learn about any one variable. Better-controlled wearer trials are needed but, unfortunately, most manufacturers would not publish such studies since the information can be commercially very valuable.

The future of PALs holds great promise. In spite of their limitations, PALs are the best choice for a large share of the presbyopic population. With adequate patient education, proper fitting and a sound choice of lens, the success rate for PAL correction of presbyopia is very high and will only get better.

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